

V-Trace

DESCRIBE → SEARCH → CONFIRM → RE-IDENTIFY

Multi-Attribute Vehicle Re-Identification System

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Pattern Recognition Systems

The Problem

A plate is powerful evidence — but CCTV does not always give us a readable plate.

01

CLEAR PLATE

- Strong identifier
- Useful when the plate is readable.

02

PARTIAL PLATE

- Still useful evidence
- But not enough to depend on alone.

03

NO PLATE

- Appearance remains
- Colour | Type | Shape | Distinctive cues

Objective

Search and re-identify a target vehicle across multiple camera recordings using all available visual evidence.

V-Trace ranks potential matches for human review — it does not claim definitive forensic identity.

How the User Uses V-Trace

The user starts with what they know — not by manually finding the vehicle in the footage.

1 UPLOAD

Camera A.mp4
Camera B.mp4
Camera C.mp4

2 DESCRIBE

White SUV
Plate: SGX12??
Rear dent
Color: Not available

3 SEARCH

#1 91%
#2 84%
#3 72%

4 CONFIRM

User selects the most plausible candidate

5 RE-IDENTIFY

Other videos
+ frames
+ timestamps

DESCRIPTION-TO-VEHICLE RETRIEVAL

VEHICLE-TO-VEHICLE Re-ID

Project Scope

A focused proof-of-concept for multi-video vehicle search and Re-ID.

IN SCOPE

- ✓ Uploaded CCTV / traffic videos
- ✓ Vehicle detection + tracking
- ✓ Colour / type / plate / appearance
- ✓ Description-based candidate search
- ✓ Cross-video Re-ID
- ✓ Ranked results + frames / timestamps
- ✓ Streamlit application

OUT OF SCOPE

- × Operational live CCTV connection
- × Vehicle-owner / registration records
- × Facial recognition of occupants
- × Definitive forensic identification
- × Training large vision models from scratch
- × Guaranteed detection of every dent / modification

MVP: recorded-video proof of concept → future extension: live camera streams

Data Required / Available

Benchmark data measures Re-ID performance; video clips demonstrate the complete software workflow.

BENCHMARK | VeRi-776

50K+ vehicle images
776 vehicle identities
20 cameras

Used for:
- known cross-camera identities
- Re-ID evaluation
- vehicle attributes / metadata

VIDEO DEMONSTRATION

Short CCTV / traffic-style clips
Multiple uploaded videos

Used for:
- vehicle detection
- within-video tracking
- description search
- end-to-end application demo

Benchmark = “Does the Re-ID method work?” | Videos = “Does the software workflow work?”

System Design

Raw videos become searchable vehicle tracks, then ranked cross-video sightings.

VIDEOS

Uploaded camera footage

DETECT + TRACK

YOLO + tracker → vehicle tracks

REPRESENT

CNN appearance
Colour • Type • Plate

INDEX + SEARCH

User description → ranked candidates

CONFIRM

Selected candidate becomes visual query

Re-ID + FUSION

Compare other tracks → ranked sightings

Tracklet / vehicle track = observations of one tracked vehicle while it is visible in one video

Technique Groups

Each technique group has a distinct role in the same end-to-end pipeline.

1 SUPERVISED / UNSUPERVISED

Supervised: same-vehicle vs different-vehicle matching
Unsupervised: SOM / clustering of vehicle embeddings

2 MACHINE / DEEP LEARNING

Pretrained vehicle detection
CNN / Re-ID appearance embeddings
Optional plate OCR

3 HYBRID / ENSEMBLE

Random Forest / weighted fusion combines:
appearance + colour + type + plate + track evidence

4 INTELLIGENT SENSING / SENSE-MAKING

Raw video → detections → tracks → descriptors
→ candidates → confirmed target → sightings

Why Combine Multiple Cues?

Every source of evidence fails differently.

CUE	STRENGTH	LIMITATION
PLATE	Strong identity	Blur / angle / occlusion
COLOUR	Simple & robust	Many vehicles share it
TYPE	Good candidate filter	Not unique
APPEARANCE	Rich visual detail	Viewpoint / lighting
TRACKING	Multi-frame evidence	ID switches / fragmentation

COMPLEMENTARY EVIDENCE → STRONGER, MORE ROBUST RANKING

Outcome & Evaluation

The goal is not just a successful demo — each key claim is tested.

1 RETRIEVAL

Can the correct vehicle be ranked highly?

Rank-1
Rank-5
mAP

2 FUSION

Does combining evidence help?

Appearance only
vs Weighted Fusion
vs Random Forest

3 PLATE ROBUSTNESS

What happens when plate evidence fails?

Clear → Partial → Blurred → Hidden

Hypothesis: multi-evidence retrieval should degrade more gracefully when one cue becomes unavailable.

Progress & Next Steps

The project is defined; implementation proceeds stage-by-stage so each component can be tested independently.

DONE

- ✓ Problem & two-stage workflow defined
- ✓ Scope and limitations fixed
- ✓ Primary dataset identified
- ✓ Four technique groups mapped
- ✓ Candidate models / tools shortlisted
- ✓ Evaluation plan defined

NEXT

Detection + Tracking

↓

Vehicle Index

↓

Stage 1 Search

↓

Cross-video Re-ID

↓

Fusion + Evaluation

↓

Streamlit Integration

Thank you!

In the preparation of this proposal, AI-based tools including Claude, ChatGPT, and Gemini were used in a supporting capacity to assist with tasks such as drafting and refining text, organizing content, and providing general research and reference support. All ideas, analysis, technical design decisions, and conclusions presented in this proposal are the work of the project team, with AI tools used solely to support and streamline the writing and preparation process.